

Lab Exercise 14

TCL COMMANDS – COMMIT, SAVEPOINT, ROLLBACK

AIM

To implement **COMMIT**, **SAVEPOINT**, and **ROLLBACK** commands on the `Vegetable` table.

PROCEDURE

1. Create the `Vegetable` table and insert records.
2. Start a transaction.
3. Update a vegetable record and create **SAVEPOINT A**.
4. Insert another vegetable and create **SAVEPOINT B**.
5. Insert another vegetable and create **SAVEPOINT C**.
6. Use **ROLLBACK TO** to return to a particular savepoint.
7. Use **COMMIT** to permanently save the remaining changes.
8. Use **SELECT** to verify the changes.

3. UPDATE and SAVEPOINT A

```
START TRANSACTION;
```

```
UPDATE Vegetable  
SET VegName = 'Beetroot'  
WHERE VegID = 5;
```

```
SAVEPOINT A;
```

Check:

```
SELECT * FROM Vegetable;
```

```
mysql> SELECT * FROM Vegetable;
```

VegID	VegName	Category	Price	Quantity
1	Tomato	Vegetable	30.50	100
2	Potato	Vegetable	25.00	200
3	Carrot	Root	40.00	80
4	Onion	Vegetable	35.00	150
5	Beetroot	Vegetable	45.50	90
6	Cabbage	Leafy	50.00	70
10	Spinach	Leafy	28.00	60
11	Beetroot	Root	42.00	50

```
8 rows in set (0.00 sec)
```

4. INSERT and SAVEPOINT B

```
INSERT INTO Vegetable  
VALUES (6, 'Cabbage', 'Leafy', 50.00, 70);
```

```
SAVEPOINT B;
```

Check:

```
SELECT * FROM Vegetable;
```

```
mysql> SELECT * FROM Vegetable;
```

VegID	VegName	Category	Price	Quantity
1	Tomato	Vegetable	30.50	100
2	Potato	Vegetable	25.00	200
3	Carrot	Root	40.00	80
4	Onion	Vegetable	35.00	150
5	Beetroot	Vegetable	45.50	90
6	Cabbage	Leafy	50.00	70
10	Spinach	Leafy	28.00	60
11	Beetroot	Root	42.00	50

8 rows in set (0.00 sec)

5. INSERT and SAVEPOINT C

```
INSERT INTO Vegetable  
VALUES (7, 'Spinach', 'Leafy', 28.00, 60);
```

```
SAVEPOINT C;
```

Check:

```
SELECT * FROM Vegetable;
```

```
mysql> SELECT * FROM Vegetable;
```

VegID	VegName	Category	Price	Quantity
1	Tomato	Vegetable	30.50	100
2	Potato	Vegetable	25.00	200
3	Carrot	Root	40.00	80
4	Onion	Vegetable	35.00	150
5	Beetroot	Vegetable	45.50	90
6	Cabbage	Leafy	50.00	70
7	Spinach	Leafy	28.00	60
10	Spinach	Leafy	28.00	60
11	Beetroot	Root	42.00	50

```
9 rows in set (0.00 sec)
```

6. Rollback to SAVEPOINT B

ROLLBACK TO B;

Then:

SELECT * FROM Vegetable;

```
mysql> SELECT * FROM Vegetable;
```

VegID	VegName	Category	Price	Quantity
1	Tomato	Vegetable	30.50	100
2	Potato	Vegetable	25.00	200
3	Carrot	Root	40.00	80
4	Onion	Vegetable	35.00	150
5	Beetroot	Vegetable	45.50	90
6	Cabbage	Leafy	50.00	70
10	Spinach	Leafy	28.00	60
11	Beetroot	Root	42.00	50

```
8 rows in set (0.00 sec)
```

7. COMMIT

COMMIT;

```
mysql> SELECT * FROM Vegetable;
```

VegID	VegName	Category	Price	Quantity
1	Tomato	Vegetable	30.50	100
2	Potato	Vegetable	25.00	200
3	Carrot	Root	40.00	80
4	Onion	Vegetable	35.00	150
5	Beetroot	Vegetable	45.50	90
6	Cabbage	Leafy	50.00	70
10	Spinach	Leafy	28.00	60
11	Beetroot	Root	42.00	50

```
8 rows in set (0.00 sec)
```